

Deriving the Sum of Powers Formulas Until the Third Power with Matrices: An Introduction to Sum of Powers

George Papakostas

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Abstract

The goal of this paper is to prove the sum of power formulas $\sum_{i=1}^n i^p$ up to $p=3$ with arguments that remain simultaneously intuitive and rigorous. I.e., Operations on matrices shall be defined and used as a convention to facilitate understanding. The paper is an introductory learning opportunity to understand the derivation of basic summation formulas. This approach is inclusive to all readers with a basic understanding of summation and matrix notation and algebra. The paper is cumulative, beginning with a definition and building off previously proven summation formulas to solve proceeding ones.

Definition 1.1. Let M be an arbitrary matrix. Then $Sum(M)$ is the summation of all entries in the matrix.

Theorem 1.2 (Sum of first n natural numbers).

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

Proof. Using the matrix sum definition, we can rewrite the sum of natural numbers as the following $n \times n$ matrix:

$$\sum_{i=1}^n i = Sum \begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 1 & 0 & \dots & 0 \\ 1 & 1 & 1 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & 1 & \dots & 1 \end{pmatrix} \quad (1)$$

The number of upper triangular 0 entries is equal to the $(n-1)$ th triangular number, since the first row contains $n-1$ zeros and every proceeding row decreases by one until the n th row. Therefore, if we add $\sum_{i=1}^{n-1} i$ to the matrix, it will contain n^2 entries of 1. Therefore:

$$\sum_{i=1}^n i + \sum_{i=1}^{n-1} i = n^2 \quad (2)$$

Essentially, the sum of the n th and the $(n-1)$ th triangular number is equal to n^2 for all natural numbers n . This fact will also be useful later on to derive the sum of squared natural numbers formula. To conclude:

$$\sum_{i=1}^n i + \sum_{i=1}^{n-1} i = 2 \sum_{i=1}^{n-1} i + n = n^2$$

$$\sum_{i=1}^{n-1} i = \frac{n(n-1)}{2}$$

$$\therefore \sum_{i=1}^n i = \frac{n(n+1)}{2}$$

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Corollary 1.3 (Sum of first n odd numbers).

$$\sum_{i=1}^n 2i - 1 = n^2$$

Proof. Using $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ from proposition 1, we can extend the sum to $2n-1$ and subtract all of the even terms to produce the sum of the first n odd natural numbers. There must be $n-1$ even numbers since the last term is odd. Therefore:

$$\begin{aligned} \sum_{i=1}^{2n-1} i - 2 \sum_{i=1}^{n-1} i &= \sum_{i=1}^n 2i - 1 \\ \frac{(2n-1)(2n)}{2} - 2 \left[\frac{(n-1)(n)}{2} \right] &= \sum_{i=1}^n 2i - 1 \\ (2n-1)(n) - (n-1)n &= \sum_{i=1}^n 2i - 1 \\ \therefore n^2 &= \sum_{i=1}^n 2i - 1 \end{aligned}$$

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Theorem 1.4 (Sum of first n squared natural numbers).

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Proof. As earlier, we will introduce the matrix sum definition to facilitate the arguments:

$$\sum_{i=1}^n i^2 = \text{Sum} \begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 2 & 2 & 0 & 0 & \dots & 0 \\ 3 & 3 & 3 & 0 & \dots & 0 \\ 4 & 4 & 4 & 4 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ n & n & n & n & \dots & n \end{pmatrix} \quad (3)$$

This is an $n \times n$ matrix, and the number of entries in a row is equal to the row's number for all rows. Therefore, from (1), the number of entries which are zero is $\sum_{i=1}^{n-1} i$. Observe that every row begins at 1 and increases by 1 until the n th row.

Consider the n th column. There are $n-1$ zeros. If we add the $(n-1)$ th triangular number, the column will transform to $\sum_{i=1}^n i$. The next row contains $n-2$ zeros, which implies that if we add the $(n-2)$ th triangular number, it will transform to $\sum_{i=1}^n i$. This pattern holds until the second column. Therefore, if we add the sum of triangular numbers up until $n-1$, the matrix will transform the following way:

$$\sum_{i=1}^n i^2 + \sum_{i=1}^{n-1} \frac{i(i+1)}{2} = \text{Sum} \begin{pmatrix} 1 & \mathbf{1} & \mathbf{1} & \mathbf{1} & \dots & \mathbf{1} \\ 2 & 2 & \mathbf{2} & \mathbf{2} & \dots & \mathbf{2} \\ 3 & 3 & 3 & \mathbf{3} & \dots & \mathbf{3} \\ 4 & 4 & 4 & 4 & \dots & \mathbf{4} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ n & n & n & n & \dots & \mathbf{n} \end{pmatrix} \quad (4)$$

Since this is an $n \times n$ matrix and every column's sum are equal to $\sum_{i=1}^n i$, the sum of the n columns must be equal to $n \sum_{i=1}^n i$. We conclude:

$$\sum_{i=1}^n i^2 + \sum_{i=1}^{n-1} \frac{i(i+1)}{2} = \frac{n^2(n+1)}{2} \quad (5)$$

Recall that from (2), we showed:

$$n^2 = \sum_{i=1}^n i + \sum_{i=1}^{n-1} i = 2 \sum_{i=1}^{n-1} (i) + n$$

If we take the sum from $n=1$ to k of both sides, we obtain:

$$\sum_{n=1}^k n^2 = 2 \sum_{n=1}^{k-1} \left(\frac{n(n+1)}{2} \right) + \frac{k(k+1)}{2}$$

Notice that we have a new representation of $\sum_{i=1}^n i^2$. If we add $\sum_{i=1}^n i^2$ to both sides of the equation, then:

$$3 \sum_{i=1}^n i^2 = 2 \sum_{i=1}^n i^2 + 2 \sum_{i=1}^{n-1} \frac{i(i+1)}{2} + \frac{n(n+1)}{2}$$

From (5), substituting $n^2(n+1) = 2 \sum_{i=1}^n i^2 + 2 \sum_{i=1}^{n-1} \frac{i(i+1)}{2}$:

$$3 \sum_{i=1}^n i^2 = n^2(n+1) + \frac{n(n+1)}{2}$$

By factoring $n(n+1)$ and isolating $\sum_{i=1}^n i^2$, we arrive at the desired formula:

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

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Corollary 1.5 (Sum of first n even squared natural numbers).

$$\sum_{i=1}^n (2i)^2 = \frac{2n(n+1)(2n+1)}{3}$$

Proof. By the exponent law, $(2i)^2=4i^2$:

$$\sum_{i=1}^n (2i)^2 = 4 \sum_{i=1}^n i^2 = \frac{2n(n+1)(2n+1)}{3}$$

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Corollary 1.6 (Sum of first n odd squared natural numbers).

$$\sum_{i=1}^n (2i-1)^2 = \frac{n(4n^2-1)}{3}$$

Proof

$$\sum_{i=1}^n (2i-1)^2 = \sum_{i=1}^n 4i^2 - 4i + 1$$

By commutativity of addition, the sum can be represented as:

$$\begin{aligned} \sum_{i=1}^n 4i^2 - 4i + 1 &= 4 \sum_{i=1}^n i^2 - 4 \sum_{i=1}^n i + \sum_{i=1}^n 1 = \frac{2n(n+1)(2n+1)}{3} - \frac{4n(n+1)}{2} + n \\ &= \frac{n[4(n+1)(2n+1) - 12(n+1) + 6]}{6} = \frac{n[4n^2 + 6n + 2 - 6n - 6 + 3]}{3} \\ &= \frac{n(4n^2-1)}{3} \end{aligned}$$

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Corollary 1.7 (Sum of first n triangular numbers).

$$\sum_{i=1}^n \frac{i(i+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

Proof. Recall from (5) that:

$$\begin{aligned} \sum_{i=1}^n i^2 + \sum_{i=1}^{n-1} \frac{i(i+1)}{2} &= \frac{n^2(n+1)}{2} \\ \frac{n(n+1)(2n+1)}{6} + \sum_{i=1}^n \frac{i(i+1)}{2} &= \frac{n^2(n+1)}{2} + \frac{n(n+1)}{2} \\ \sum_{i=1}^n \frac{i(i+1)}{2} &= \frac{3n^2(n+1) + 3n(n+1) - n(n+1)(2n+1)}{6} \end{aligned}$$

$$\sum_{i=1}^n \frac{i(i+1)}{2} = \frac{n(n+1)[3n+3-2n-1]}{6}$$

$$\therefore \sum_{i=1}^n \frac{i(i+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

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Theorem 1.8 (Sum of first n cubed natural numbers).

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Proof. The main idea is to create an equation involving $\sum_{i=1}^n i^3$ with sums of degree less than 3 since they have already been derived. The methods from (2) and (4) can be used, except now the matrix representation will have squared entries:

$$\sum_{i=1}^n i^3 = \text{Sum} \begin{pmatrix} 1^2 & 0 & 0 & 0 & \dots & 0 \\ 2^2 & 2^2 & 0 & 0 & \dots & 0 \\ 3^2 & 3^2 & 3^2 & 0 & \dots & 0 \\ 4^2 & 4^2 & 4^2 & 4^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ n^2 & n^2 & n^2 & n^2 & \dots & n^2 \end{pmatrix}$$

It is obvious that this equation is true since for every row k there exists k multiples of k^2 which is equivalent to k^3 . Analogous to the argument in (4), if we add $\sum_{j=1}^{n-1} \sum_{i=1}^j i^2 = \sum_{i=1}^{n-1} \frac{i(i+1)(2i+1)}{6}$, the matrix will have each of its rows equal to $\sum_{i=1}^n i^2$. I.e.,

$$\sum_{i=1}^n i^3 + \sum_{i=1}^{n-1} \frac{i(i+1)(2i+1)}{6} = \text{Sum} \begin{pmatrix} 1^2 & 1^2 & 1^2 & 1^2 & \dots & 1^2 \\ 2^2 & 2^2 & 2^2 & 2^2 & \dots & 2^2 \\ 3^2 & 3^2 & 3^2 & 3^2 & \dots & 3^2 \\ 4^2 & 4^2 & 4^2 & 4^2 & \dots & 4^2 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ n^2 & n^2 & n^2 & n^2 & \dots & n^2 \end{pmatrix} \quad (6)$$

$$\Leftrightarrow \sum_{i=1}^n i^3 + \sum_{i=1}^{n-1} \frac{i(i+1)(2i+1)}{6} = n \sum_{i=1}^n i^2$$

Isolating for $\sum_{i=1}^n i^3$:

$$8 \sum_{i=1}^{n-1} (i^3) + 6n^3 + \frac{n(n-1)(2n-1)}{2} + \frac{n(n-1)}{2} = n^2(n+1)(2n+1)$$

$$8 \sum_{i=1}^{n-1} i^3 = \frac{2n^2(n+1)(2n+1) - 12n^3 - 2n^2(n-1)}{2}$$

$$8 \sum_{i=1}^{n-1} i^3 = n^2(2n^2 - 4n + 2)$$

$$\sum_{i=1}^{n-1} i^3 = \frac{n^2(n-1)^2}{4} = \left(\frac{n(n-1)}{2}\right)^2$$

$$\therefore \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

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Remark 1.9

It is quite surprising that the sum of the first n natural cubed numbers is equal to the sum of the first n natural numbers all squared. This implies $\sum_{i=1}^n i^3$ is a perfect square for all n . In fact this is precisely why a proof by induction would be a great method since there is a clear pattern that can be observed.

Without loss of generality, the process by which we solved for proceeding sums according to (2), (4), and (6) can be represented as:

$$\sum_{i=1}^n i^p + \sum_{i=1}^{n-1} \sum_{j=1}^i j^{p-1} = n \sum_{i=1}^n i^{p-1} \quad (7)$$

Definition 1.10

Let $p(n)$ be the polynomial in terms of n equal to $\sum_{i=1}^n i^k$ for any natural number k . Then:

$$\text{Max}(p(n)) := \text{highest order term of } p(n).$$

Theorem 1.11

Let $p(n)$ be a polynomial in terms of n equal to $\sum_{i=1}^n i^k$ for some natural number k . Then the highest order term of $p(n)$ is equal to $\frac{1}{k+1} n^{k+1}$.

Proof. We will use induction since the goal is to show $\text{Max}(p(n)) = \frac{1}{k+1} n^{k+1}$ for all natural numbers k . additionally we will consider the equation from (7). We know that:

$$\sum_{i=1}^n i = \frac{n^2}{2} + \frac{n}{2}$$

$$\text{Max}\left(\sum_{i=1}^n i\right) = \frac{1}{2}n^2$$

This first case satisfies $\text{Max}\left(\sum_{i=1}^n i^k\right) = \frac{1}{k+1}n^{k+1}$. We shall prove the proceeding case before we determine all cases of k. Using (7):

$$\begin{aligned} \sum_{i=1}^n i^2 + \sum_{i=1}^{n-1} \sum_{j=1}^i j &= n \sum_{i=1}^n i \\ \Leftrightarrow \sum_{i=1}^n i^2 + \sum_{i=1}^{n-1} \frac{i^2}{2} + \sum_{i=1}^{n-1} \frac{i}{2} &= n \left(\frac{n^2}{2} + \frac{n}{2} \right) \end{aligned}$$

Adding $\frac{n^2}{2}$ to both sides of the equation:

$$\sum_{i=1}^n i^2 + \sum_{i=1}^n \left(\frac{i^2}{2} \right) + \sum_{i=1}^{n-1} \frac{i}{2} = n \left(\frac{n^2}{2} + \frac{n}{2} \right) + \frac{n^2}{2}$$

Multiplying both sides of the equation by 2:

$$\begin{aligned} 3 \sum_{i=1}^n i^2 &= n^3 + 2n^2 \cdot \sum_{i=1}^{n-1} i \\ \Leftrightarrow \sum_{i=1}^n i^2 &= \frac{1}{3}n^3 + \frac{2}{3}n^2 \cdot \frac{1}{3} \sum_{i=1}^{n-1} i \end{aligned}$$

Note that the highest order of $\sum_{i=1}^{n-1} i$ is less than 3, which implies that n^3 and $\sum_{i=1}^{n-1} i$ are linearly independent. Therefore:

$$\text{Max}\left(\sum_{i=1}^n i^2\right) = \frac{1}{3}n^3$$

WLOG, from (7)

$$\sum_{i=1}^n i^k + \sum_{i=1}^{n-1} \sum_{j=1}^i j^{k-1} = n \sum_{i=1}^n i^{k-1} \quad (8)$$

Observe that on the right hand side of the equation:

$$\text{Max}\left(n \sum_{i=1}^n i^{k-1}\right) = \frac{1}{k}n^{k+1} \quad (9)$$

Note that this conclusion is valid since $k=3 \Rightarrow \text{Max}(\sum_{i=1}^n i^{k-1}) = \text{Max}(\sum_{i=1}^n i^2) = \frac{1}{k} n^k$. On the left hand side of the equation, it would be helpful to simplify $\sum_{i=1}^{n-1} \sum_{j=1}^i j^{k-1}$:

$$\text{Max} \left(\sum_{j=1}^i j^{k-1} \right) = \frac{1}{k} i^k$$

We shall define $f(n)$ to be:

$$\text{Max} \left(\sum_{j=1}^i j^{k-1} \right) + f(n) = \sum_{j=1}^i j^{k-1}$$

Therefore,

$$\sum_{i=1}^{n-1} \sum_{j=1}^i j^{k-1} = \sum_{i=1}^{n-1} \frac{1}{k} i^k + \sum_{i=1}^{n-1} f(n)$$

Remark 1.12 Notice that the largest exponent of $\sum_{i=1}^{n-1} f(n) \leq k$ since the exponent of $\text{Max}(\sum_{j=1}^i j^{k-1})$ is greater the exponent of $\text{Max}(f(n))$.

Substituting into (8):

$$\sum_{i=1}^n i^k + \frac{1}{k} \sum_{i=1}^{n-1} i^k + \sum_{i=1}^{n-1} f(n) = n \sum_{i=1}^n i^{k-1}$$

Adding $\frac{1}{k} n^k$ to both sides of the equation, then multiplying by k , we obtain:

$$\begin{aligned} k \sum_{i=1}^n i^k + \sum_{i=1}^n i^k + k \sum_{i=1}^{n-1} f(n) &= nk \sum_{i=1}^n i^{k-1} + n^k \\ (k+1) \sum_{i=1}^n i^k &= nk \sum_{i=1}^n i^{k-1} + n^k - k \sum_{i=1}^{n-1} f(n) \end{aligned}$$

Recall that $\text{Max}(n \sum_{i=1}^n i^{k-1}) = \frac{1}{k} n^{k+1}$. From remark 1.12, we can conclude that the largest exponent terms of both n^k and $k \sum_{i=1}^{n-1} f(n)$ are less than $k+1$, which implies that $nk \sum_{i=1}^n i^{k-1}$ is linearly independent from the other terms. Therefore, using (9):

$$\text{Max} \left((k+1) \sum_{i=1}^n i^k \right) = \text{Max} \left(nk \sum_{i=1}^n i^{k-1} \right) = n^{k+1}$$

$$\therefore \text{Max} \sum_{i=1}^n i^k = \frac{1}{k+1} n^{k+1}$$

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Remark 1.13

From these findings, the leading term of a sum of powers $\sum_{x=1}^n x^k$ is approximately equal to the area under the continuous curve x^k , for all natural numbers k . This makes intuitive sense since $\sum_{x=1}^n x^k$ is the area under the curve of $f(x) = x^k$ using rectangles with a width of 1. Thus:

$$\sum_{x=1}^n x^k \approx \int_0^n x^k$$

Conclusion

The following summation formulas were proven cumulatively. Matrix sum representations facilitated understanding of the arguments for each proof. We began by showing that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ then used it to prove that $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$. Then, these facts are used to prove $\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$. Finally, the method of solving each power sum is generalized to prove a theorem that the highest order term in a power sum $\sum_{i=1}^n i^k$ is equal to $\frac{1}{k+1} n^{k+1}$.

The goal of this paper is to demonstrate methods of deriving the first three summation formulas in an intuitive, yet rigorous method. Generalizing the results of this paper to $\sum_{i=1}^n i^k$ for any natural number k is a project for future research.